

Game plan:

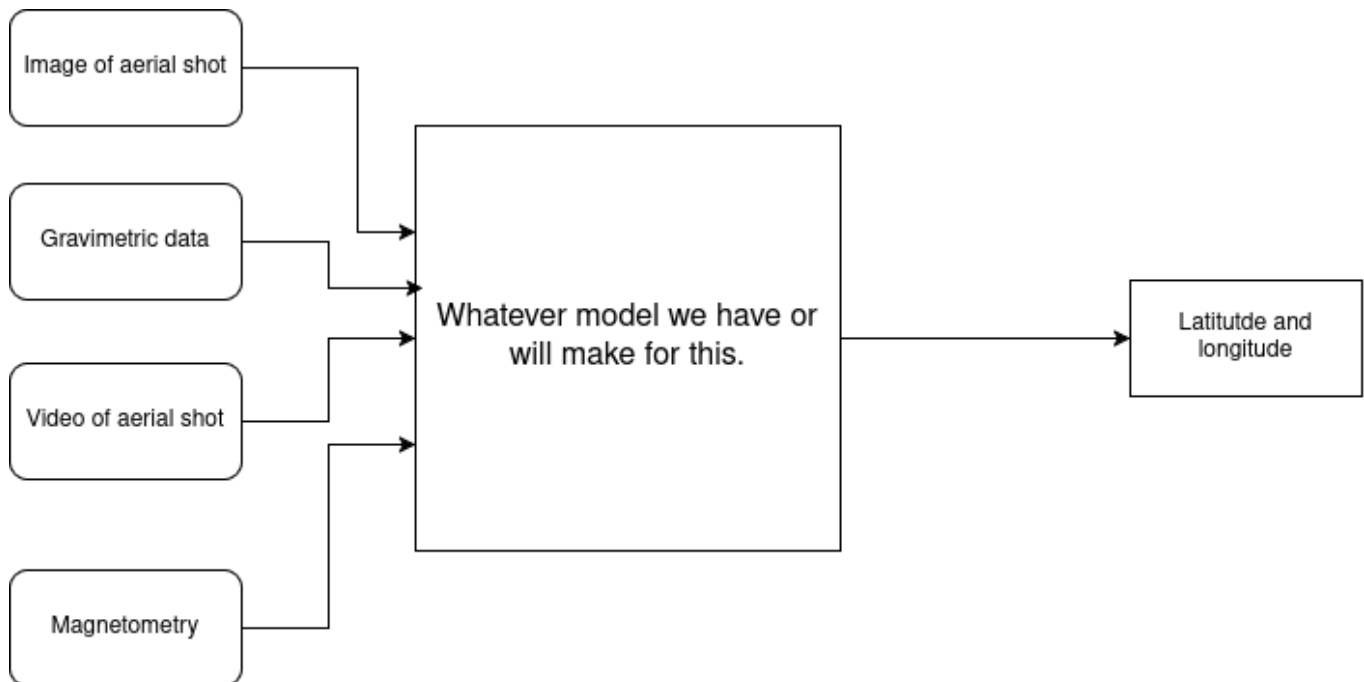
- This sem get system working
- Next sem make it fast and power efficient.

Objectives

- Scene Aided navigation
- GPS Denied system and any kind of GNSS
- We can have magnetometers/gravity measurement but have to evaluate feasibility.
- TERCOM and DSMAC Cold war stuff most promising for now (and obviously proven.)

Current Objectives:

- Replicate any research paper/any code that deals with this in simulation.
- Since we aren't going to use drones initially, to prove feasibility, we provide aerial shots to our black box and get solution.
- measured in terms of input and output:



Other stuff that exists

Method	What you measure	Reference map/database	GPS-denied?	Passive?	Do we like this one
Gravity matching	(g), gravity anomalies	Gravity map	Yes	Yes	yes
Gravity-gradient navigation	($\partial g / \partial x, y, z$)	Gravity-gradient map	Yes	Yes	very much
Magnetic anomaly navigation	Magnetic field/anomalies	Magnetic map	Yes	Yes	nah but could be useful since it's mountains ranges and stuff that are looking at.
Terrain matching (TERCOM)	Elevation	DEM/terrain map	Yes	Mostly	Very nice
Bathymetric matching	Seafloor depth	Bathymetric map	Yes	Yes	Nope. not all.
Celestial navigation	Stars/Sun	Astronomical catalog	Yes	Yes	age old. could be useful at night but our accuracy purposes maybe not but maybe we could do an initial reading or something make our other methods more powerful And it works on in the night
Visual landmark matching	Camera imagery	Satellite/landmark map	Yes	Yes	well. If it's AI visual model, then

Method	What you measure	Reference map/database	GPS-denied?	Passive?	Do we li this one
					will automati happen.
Lidar map matching	3D geometry	LiDAR/3D map	Yes	Yes	not in ou scope
Radar map matching	Radar reflectivity/geometry	Radar map	Yes	Yes	not in sci
Wi-Fi fingerprinting	RSSI/CSI	Wi-Fi fingerprint map	Yes	No	how tf we there be so high u the mountair no.
Cellular fingerprinting	Tower signals	Cellular database	Yes	No	urban navigatic of course literal ligh beams th are alwa present & dont mov all. All weather. good for triangula they eve identify themselv Approver
LEO signals of opportunity	Doppler / timing	Satellite ephemerides	Yes	No	nope.
Ambient RF fingerprinting	Radio spectrum	RF fingerprint map	Yes	No	Did not explore, maybe w it.
Acoustic matching	Sound/echo characteristics	Acoustic map	Yes	Sometimes	nope. no working t high up.
Thermal fingerprinting	IR temperature pattern	Thermal map	Yes	Yes	could be extra dat

Method	What you measure	Reference map/database	GPS-denied?	Passive?	Do we like this one
					point idk.
Atmospheric-field matching	Pressure/temp/wind	Atmospheric model	Partly	Yes	nope. too volatile right now
Quantum inertial sensing	Acceleration/rotation	No map necessarily	Yes	Yes	This is interesting. I don't have another idea. Basically just track initial position & direction, where we start the mission, at and have good sensors onboard to detect acceleration, direction, rotation, direction, thrust, everything, and then we can probably just use physics & capture the final state we are in. this could probably work very, very well.

Stuff we need to do rn

The below paper basically works for our model, so we first try to replicate it:

- <https://github.com/hmgoforth/gps-denied-uav-localization>
- <https://publications.ri.cmu.edu/gps-denied-uav-localization-using-pre-existing-satellite-imagery>

Pre-existing code

1. https://github.com/snktshrma/ngps_flight — Real-time CLIP-style matching, ~3m RMSE, runs on Jetson Orin.
2. <https://github.com/YFS90/GNSS-Denied-UAV-Geolocalization> — Terrain-weighted optimization using satellite maps and DEM data, sub-7m.
3. <https://github.com/hmgoforth/gps-denied-uav-localization> — Fine-tuned VGG16 deep features matching drone frames to satellite tiles.
4. <https://github.com/codebra721/CLIP-UAV-localization> — CLIP-based network matching UAV images, also recovers heading direction.
5. <https://github.com/OSUPCVLab/PCVLabDrone2021> — Real-time down-facing camera pipeline with quad-tree map retrieval, tested live.
6. <https://github.com/abhinavtrpathi95/geolocalization> — Dual/Siamese network recognizing same-scene UAV and satellite image pairs.
7. <https://github.com/sidharthmohannair/VisionUAV-Navigation> — Classical SIFT/ORB hybrid feature detection, no deep learning needed.

Research papers:

Image Based

1. <https://www.sciencedirect.com/science/article/pii/S1569843224006332> — GNSS-denied UAV geolocalization via terrain-weighted constraint optimization, sub-7m error.
2. <https://www.sciencedirect.com/science/article/pii/S2590123025041787> — CLIP-based UAV-satellite image matching, also derives heading direction.
3. <https://arxiv.org/pdf/2103.14381> — GNSS-denied geolocalization via onboard camera matching against orthophotos.
4. <https://arxiv.org/html/2503.10692v1> — AnyVisLoc benchmark, multi-city, multi-altitude UAV localization dataset comparison.
5. <https://arxiv.org/pdf/2509.13795> — SWA-PF particle filter for memory-efficient 4-DoF GNSS-denied UAV localization.
6. <https://www.sciencedirect.com/science/article/abs/pii/S0924271624002041> — Coarse-to-fine visual geo-localization handling oblique-view UAV imagery robustly.
7. <https://www.sciencedirect.com/science/article/abs/pii/S0957417424029312> — Hierarchical localization method cutting UAV positioning error significantly.

8. <https://dl.acm.org/doi/10.1145/3722405.3722421> — Absolute visual localization via deep-learning feature matching against satellite maps.
9. <https://polen.itu.edu.tr/items/1fe1e872-7cea-44d8-a8de-339e4587bee6> — Thesis combining visual odometry with satellite image matching corrections.
10. <https://www.sciencedirect.com/science/article/pii/S1569843224006332> — Terrain-weighted constraint optimization achieving sub-7m GNSS-denied accuracy.
11. <https://www.sciencedirect.com/science/article/pii/S2590123025041787> — CLIP-based aerial-satellite matching network, also derives heading direction.
12. <https://arxiv.org/pdf/2103.14381> — Onboard camera matching against orthophotos for absolute geolocalization.

Gravimetric

1. A Review of Key Technologies in Gravity Matching Navigation — Comprehensive 2026 review covering sensors, maps, algorithms, and integration.
<https://www.mdpi.com/1424-8220/26/13/4208>
2. Gravity-Aided Navigation Using Viterbi Map Matching Algorithm — Uses HMM and Viterbi decoding for gravity-based trajectory estimation.
<https://arxiv.org/abs/2204.10492>
3. Gravity-Aided Navigation Using Viterbi Map Matching Algorithm — Journal version detailing probabilistic gravity-map trajectory matching techniques.
<https://doi.org/10.1017/S0373463324000250>
4. Performance Evaluation and Analysis for Gravity Matching Aided Navigation — Evaluates map resolution, sensor accuracy, and positioning performance.
<https://www.mdpi.com/1424-8220/17/4/769>
5. Gravity-Matching Algorithm Based on K-Nearest Neighbor — Applies KNN searching to match measured gravity against databases.
<https://www.mdpi.com/1424-8220/22/12/4454>
6. Gravity Matching Algorithm Based on Backtracking for Small Range Adaptation Area — Uses backtracking to improve matching within limited map regions.
<https://doi.org/10.1109/TIM.2024.3350139>
7. Reference Composite Features Database Construction Method Based on Track Segmentation — Builds gravity databases using segmented real-world measurement tracks.
<https://pubmed.ncbi.nlm.nih.gov/39909807/>
8. Gravity Matching Navigation Algorithm with Affine Transformation Based on Particle Swarm Optimization — Combines optimization and matching for improved gravity-navigation accuracy.
<https://www.sciencedirect.com/science/article/pii/S026322412600148X>

Magnetometric

1. Absolute Positioning Using the Earth's Magnetic Anomaly Field
Uses scalar magnetometer measurements and particle filtering for global positioning.
<https://doi.org/10.1002/navi.138>
2. Navigation of Ground Vehicles Using Magnetic Field Variations
Uses three-axis magnetometers and magnetic maps for vehicle position updates.
<https://doi.org/10.1002/navi.70>
3. The Vector Matching Method in Geomagnetic Aiding Navigation
Matches full magnetic vectors to improve positioning in weakly varying regions.
<https://doi.org/10.3390/s16071120>
4. Magnetic Anomalies as a Reference for Ground-speed and Map-matching Navigation
Early work exploring magnetic-field variations for ground-referenced navigation.
<https://doi.org/10.1017/S0373463300022025>
5. MagSLAM: Aerial Simultaneous Localization and Mapping Using Earth's Magnetic Anomaly Field
Performs magnetic SLAM without requiring a pre-existing magnetic anomaly map.
<https://doi.org/10.1002/navi.352>
6. Simultaneous Localization and Mapping Method for Geomagnetic Aided Navigation
Combines geomagnetic matching, SLAM, and filtering for long-scale navigation.
<https://doi.org/10.1016/j.optlastec.2018.11.014>
7. Magnetic Anomaly Navigation Using Particle Filtering and Enhanced Velocity Propagation
Combines particle filtering and IMU velocity estimation for UAV magnetic navigation.
<https://doi.org/10.1016/j.measurement.2026.121656>
8. MagNav-Based Navigation Simulation Using Magnetic Anomaly Maps for GPS-Denied Environments
Integrates quantum magnetometers, magnetic maps, INS, and extended Kalman filtering.
<https://doi.org/10.5302/J.ICROS.2026.25.0344>
9. A Survey of Magnetic-Field-Based Indoor Localization
Broad survey covering magnetic maps, fingerprinting, SLAM, filters, and neural networks.
<https://doi.org/10.3390/electronics11060864>
10. Location Fingerprint Extraction for Magnetic Field Magnitude Based Indoor Positioning
Studies magnetic fingerprint extraction and distinguishability using smartphone magnetometers.
<https://doi.org/10.1155/2016/1945695>
11. A Fine-Grained Indoor Fingerprinting Localization Based on Magnetic Field Strength and CSI

Combines magnetic fingerprints with Wi-Fi channel-state information for indoor positioning.

<https://doi.org/10.1016/j.pmcj.2017.08.003>